

Mid-Term Examination – April 2024

Programme: B.Tech (AI&DS, AI&ML, IOT)

Paper Code: AIDS204/AIML204/IOT 204

Time: 1½Hrs.

Semester: IV

Paper Name: Database Management Systems

Maximum Marks: 30

Note:

- Question No. 1 is compulsory.
- Attempt any two questions from the remaining questions.
- All questions carry equal marks.
- Only scientific calculator is allowed.

Q. No.	Question 1	Marks	CO
1(a)	What is data independence in DBMS. Briefly explain Logical and Physical data independence with the help of diagram.	[2.5]	1
1(b)	Explain various steps of Query Processing and Optimization in DBMS.	[2.5]	2
1(c)	"How does a weak and strong entity set differs from each other?" Justify your answer using real time example.	[2.5]	1
1(d)	"Referential integrity helps to maintain the consistency of a database" Justify the statement with the help of an example in RDBMS.	[2.5]	2
Question 2			
2(a)	- Consider a large multinational Bank with multiple branches across various regions globally. The Bank wants to implement a new database management system to streamline its operations and improve data management. Discuss how each of the following database models - hierarchical, network, and relational - could be applied to address the bank's needs. - Provide specific examples of how each model discussed above would handle the Bank's data structure, retrieval, and relationships between different entities within the Bank.	[5]	1
2(b)	Write short note on any two of the following concepts with suitable examples: 1) Generalization vs Specialization 2) Super key vs Candidate key 3) DBMS vs File management system	[5]	1
Question 3			
3(a)	Consider a simple schema for an employee database with the following relations: Employee (EmpID, Emp_Name, DeptID, Salary) Department (DeptID, DeptName, ManagerID) Project (ProjID, ProjName, DeptID) WorksOn (EmpID, ProjID, Hours)	[4]	2

	Write SQL queries on above relations for the each of the given queries: 1. Find the name of all the employees working for 'ABC' department. 2. Find the maximum salary of all employees. 3. Find the names of employees who work on the project named 'XYZ' 4. Retrieve Employee names starting with 'A' and ending with 'T' 5. Retrieve employee name and department id in increasing order of salary. 6. Find name of employee who work in department 'CSE' 7. Find employees earning more than 65,000 8. Retrieve the names of employees working on projects along with project names.		
3(b)	Why we use Join in DBMS? Discuss various types of inner join and outer join with suitable examples.	[6]	2
Question 4			
4(a)	Design an ER diagram for a Hospital Management System . Consider the following entities for the system: Patient, Doctor, Nurse, Department, Ward, and Appointment. A doctor can examine many patients. Clearly indicate the relationships between these entities and any attributes each entity may have. Assume the data wherever necessary. Also show the concept of Specialization in the ER diagram.	[5]	1
4(b)	Describe the role and functionality of triggers in PL/SQL. Give an example where a trigger would be beneficial and explain how to implement it while inserting or updating a record in Employee table, using PL/SQL code.	[5]	2